Bayesian Analysis of Varicella Vaccination Coverage

Project Assignment Bayesian Inference
Project 2: Birth year
2024-2025

Group 3
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Question 1:

Derive analytically the posterior of the vaccination coverage per birth year and region. Use a conjugate prior that (1) reflects no knowledge on the vaccination coverage, and (2) reflects that vaccination coverage is typically around 90% or higher. Give posterior summary measures of the vaccination coverage per birth year and region. Is the choice of the prior impacting your results?

Answer:

1.1 Analytical posterior via Beta–Binomial conjugacy

We observe for each region i and birth year j:

$$Y_{ij} \sim \text{Binomial}(N_{ij}, \pi_{ij}),$$

where Y_{ij} is the number vaccinated out of N_{ij} surveyed. We place a prior distribution on the true vaccination coverage π_{ij} and update it using Bayes Rule. Because the Beta distribution is conjugate to the Binomial, the resulting posterior is also a Beta distribution, making the analysis mathematically convenient and analytically tractable.

Why Beta Priors?

The Beta distribution is the natural conjugate prior for Binomial likelihoods. This means that if the likelihood follows a Binomial distribution and we place a Beta prior on the success probability π_{ij}, the resulting posterior will also follow a Beta distribution. This conjugacy simplifies computation and allows us to derive the posterior in closed form.
The Beta distribution is parameterized as $\text{Beta}(\alpha, \beta)$, with:

- **Mean:** $\dfrac{\alpha}{\alpha+\beta}$
- **Variance:** $\dfrac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$

We consider two types of priors to reflect different levels of prior knowledge:

- **Non-informative Prior:** $\pi_{ij} \sim \text{Beta}(1,1)$ — a uniform distribution on $[0,1]$ expressing no prior preference for any coverage rate.
- **Informative Prior:** $\pi_{ij} \sim \text{Beta}(50,2)$ — based on domain knowledge that coverage is typically high (around or above 90%). This prior has a mean of $\frac{50}{52} \approx 0.96$ and is right-skewed, assigning more weight to values near 1.

Given $Y_{ij} \sim \text{Binomial}(N_{ij}, \pi_{ij})$ and $\pi_{ij} \sim \text{Beta}(\alpha, \beta)$, the posterior is:

$$\pi_{ij} \mid Y_{ij}, N_{ij} \sim \text{Beta}(\alpha + Y_{ij}, \beta + N_{ij} - Y_{ij}).$$

From this posterior, we compute:

- **Posterior Mean:**

$$\frac{\alpha^*}{\alpha^* + \beta^*}, \quad \text{where } \alpha^* = \alpha + Y_{ij},\ \beta^* = \beta + N_{ij} - Y_{ij}$$

- **Posterior Variance:**

$$\frac{\alpha^* \beta^*}{(\alpha^* + \beta^*)^2(\alpha^* + \beta^* + 1)}$$

- **95% Equal-Tailed Credible Interval:**

$$[\text{qBeta}(0.025; \alpha^*, \beta^*),\ \text{qBeta}(0.975; \alpha^*, \beta^*)]$$

These summaries allow us to estimate coverage for each region–year combination and to evaluate how different prior assumptions influence the resulting posterior.

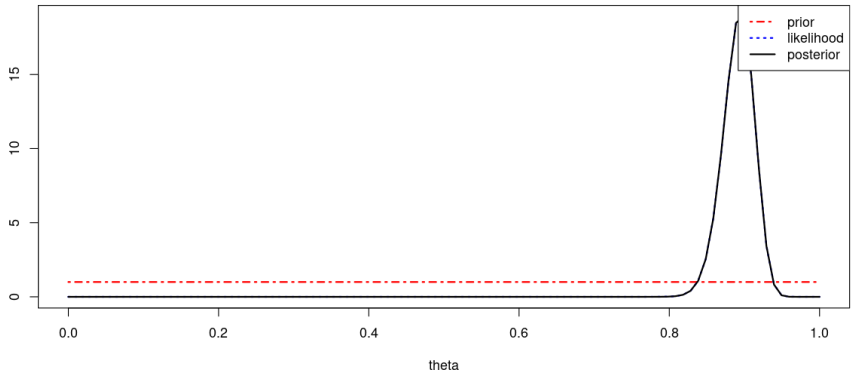


Figure 1: Posterior distribution with Non-Informative Prior Beta(1, 1)

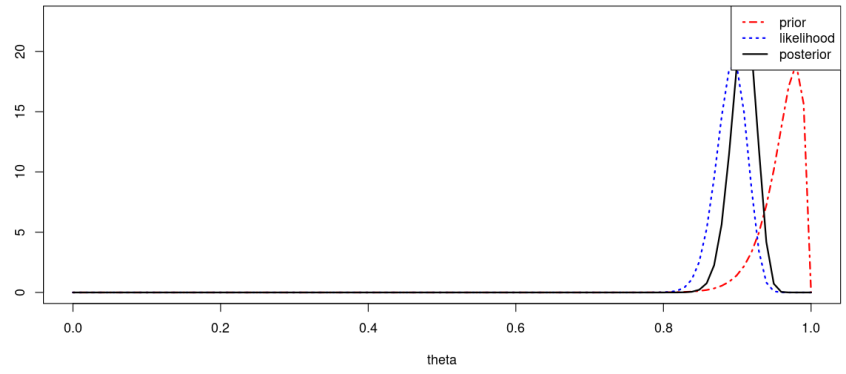


Figure 2: Posterior distribution with Informative Prior Beta(50, 2)

1.2 Summary of posterior for Georgia

Table 1: Posterior summaries of varicella vaccination coverage by birth year in Georgia under two Beta priors. (Coverage estimates for Wisconsin and Mississippi appear in Appendix A.)

Region	Birth Year	Prior Type	Posterior mean	Posterior variance	95% CI lower	95% CI upper
Georgia	2011	Informative Beta(50,2)	0.908	0.000308	0.871	0.939
Georgia	2011	Uninformative Beta(1,1)	0.895	0.000436	0.847	0.929
Georgia	2012	Informative Beta(50,2)	0.925	0.000214	0.894	0.952
Georgia	2012	Uninformative Beta(1,1)	0.919	0.000284	0.880	0.945
Georgia	2013	Informative Beta(50,2)	0.948	0.000149	0.922	0.969
Georgia	2013	Uninformative Beta(1,1)	0.946	0.000194	0.912	0.967
Georgia	2014	Informative Beta(50,2)	0.893	0.000270	0.864	0.929
Georgia	2014	Uninformative Beta(1,1)	0.887	0.000356	0.845	0.919
Georgia	2015	Informative Beta(50,2)	0.911	0.000227	0.879	0.938
Georgia	2015	Uninformative Beta(1,1)	0.902	0.000293	0.863	0.930
Georgia	2016	Informative Beta(50,2)	0.940	0.000157	0.909	0.958
Georgia	2016	Uninformative Beta(1,1)	0.931	0.000197	0.899	0.954
Georgia	2017	Informative Beta(50,2)	0.920	0.000224	0.884	0.943
Georgia	2017	Uninformative Beta(1,1)	0.908	0.000292	0.868	0.936
Georgia	2018	Informative Beta(50,2)	0.890	0.000301	0.850	0.918
Georgia	2018	Uninformative Beta(1,1)	0.872	0.000398	0.828	0.906
Georgia	2019	Informative Beta(50,2)	0.930	0.000210	0.895	0.952
Georgia	2019	Uninformative Beta(1,1)	0.920	0.000278	0.881	0.946
Georgia	2020	Informative Beta(50,2)	0.900	0.000387	0.854	0.931
Georgia	2020	Uninformative Beta(1,1)	0.878	0.000578	0.823	0.917

1.3 Interpretation

Despite using very different priors (non-informative vs informative), the posterior means differ by at most two percentage points, because each data cell has 180–330 observations. Estimated coverage stays around 87–94% with 95% intervals spanning roughly ten points. No clear increasing or decreasing trend appears once sampling variability is accounted for. In other words, our results are driven by the data, and remain consistently close to 90%, with only modest dependence on prior choice.

Question 2:

Investigate whether there is a change in the vaccination coverage over the birth years 2011–2019 using a logistic regression model

$$Y_{ij} \;\sim\; \text{Binomial}\big(N_{ij},\, \pi_{ij}\big),$$

with:

$$\text{logit}(\pi_{ij}) \;=\; \beta_0 \;+\; \beta_1 \left(\text{BirthYear}_j\right),$$

where i is the location, j is birth cohort and π_{ij} is the vaccination coverage. Assume *non-informative* priors for for the parameters to be estimated. Write and explain the code in BUGS language.

Answer:
We investigate changes in vaccination coverage across birth years (2011–2019) using a Bayesian logistic regression. The model assumes a binomial likelihood with a logit link for the centered year covariate.

BUGS Model Code

```
# Define the BUGS model
model {
  for (n in 1:Nobs) {
  # Likelihood: number vaccinated ~ Binomial(total, coverage)
    Y[n] ~ dbin(pi[n], N[n])
    logit(pi[n]) <- beta0 + beta1 * (Year[n] - 2015)
  }
# Flat (nearly  n o n informative ) priors on the regression coefficients:
  beta0 ~ dnorm(0, 1.0E-6)
  beta1 ~ dnorm(0, 1.0E-6)
}
```

First, we filter our data to the birth cohorts 2011–2019 and assemble it into `dataList`, which contains the observed counts `Y`, sample sizes `N`, the covariate `Year`, and the total number of observations `Nobs`. Next, `model.inits` provides three distinct starting values for the intercept β_0 and slope β_1, each with its own RNG seed, so that JAGS can run three independent MCMC chains. We then store the BUGS model in the string `modelString`, looping over `n in 1:Nobs` to declare the Binomial likelihood

$$Y[n] \sim \text{Binomial}\big(N[n],\, \pi[n]\big)$$

and the logistic regression link, with the year term centered at 2015:

$$\mathit{logit}\big(\pi[n]\big) = \beta_0 + \beta_1 \left(\text{Year}[n] - 2015\right).$$

We use very flat, non-informative priors:

$$\beta_0, \beta_1 \sim \text{Normal}(0, 10^6)$$

In BUGS syntax, this is written with precision as:

`dnorm(0, 1.0E-6)`

Here, precision $\tau = 1/\sigma^2$, so a small precision value (1e-6) corresponds to a large variance this ensures a very wide, flat prior that minimally influences the posterior.

Question 3:

Run the MCMC method and check convergence of the MCMC chains. Give the details on how you checked convergence.

Answer:

We implemented the Bayesian logistic regression model in JAGS and conducted Markov Chain Monte Carlo (MCMC) sampling using three chains. Multiple chains (n = 3) were employed to ensure an accurate assessment of convergence, utilizing diagnostic tools such as the Gelman-Rubin statistic. For the MCMC setup, we initialized the model with three parallel chains and applied a burn-in period of 7000 iterations to allow the chains to reach the stationary distribution before sampling. Subsequently, we drew 10000 posterior samples per chain for the parameters of interest: β_0 (intercept) and β_1 (slope for birth year).

To assess the convergence of the MCMC chains, we examined several standard diagnostic tools. One of the primary methods was the inspection of trace plots for both parameters—the intercept (β_0) and the slope (β_1)—to evaluate mixing and stability across chains. The corresponding trace plots are shown in Figure 3. The plots demonstrated good overlap and consistent behavior over time, with no discernible trends, drifts, or indications of non-convergence. This suggests that the chains had reached their stationary distributions and were sampling effectively from the posterior, supporting the conclusion that convergence was successfully achieved. In addition, posterior density plots for β_0 and β_1, also displayed in Figure 3,showed overlapping and consistent distributions across chains, further suggesting convergence.

To further convergence, we examined autocorrelation plots of the MCMC samples. As shown in Figure 4, these plots reveal low autocorrelation beyond the initial lags, indicating efficient mixing of the chains and supporting the reliability of the posterior parameter estimates.
We also applied the Gelman-Rubin diagnostic (Potential Scale Reduction Factor, PSRF) to assess convergence. For both parameters, the PSRF values were exactly 1.00—well below the commonly accepted threshold of 1.1—indicating that the between-chain and within-chain variances had stabilized and confirming convergence. The evolution of the PSRF over iterations was visualized in Figure 5, which confirms that the chains converged well before the end of the sampling period.

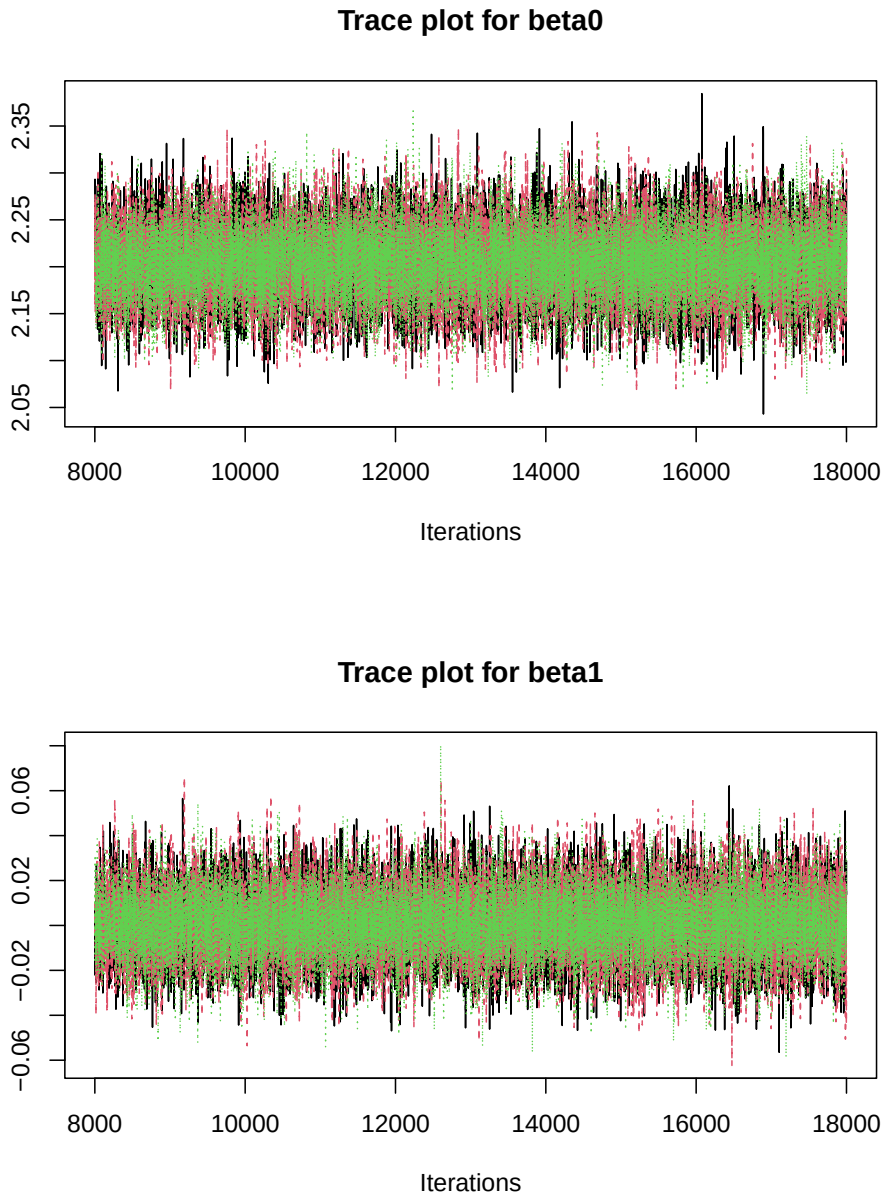


Figure 3: Trace plots for β_0 (top) and β_1 (bottom) parameters

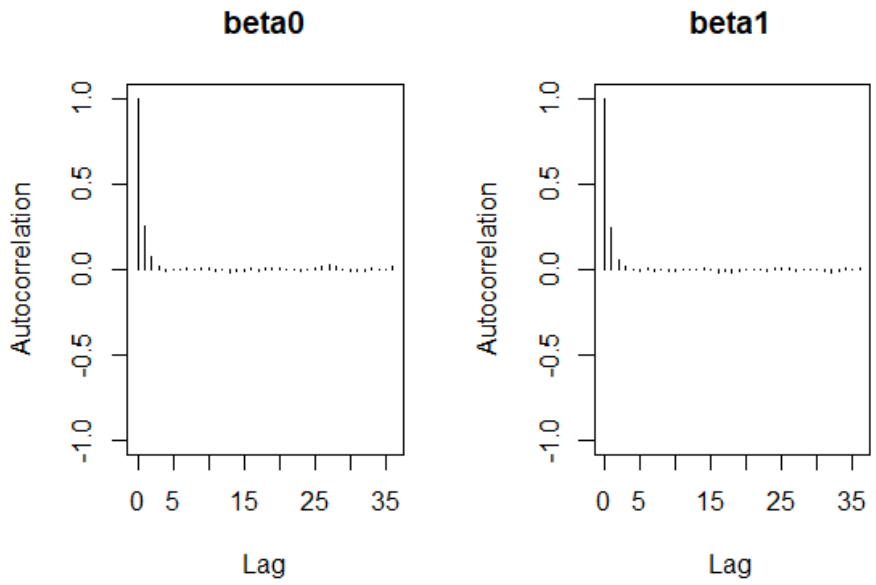


Figure 4: Autocorrelation plots for β_0 and β_1 parameters

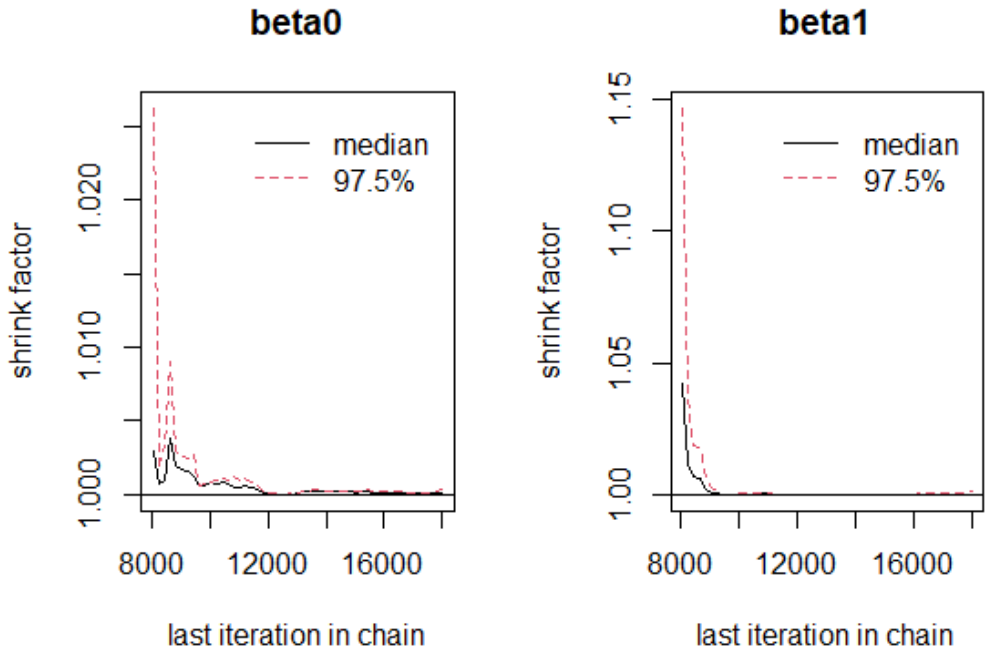


Figure 5: Gelman-Rubin diagnostic plots for β_0 and β_1 parameters

Question 4:

Make a plot of the posterior densities and give summary measures of the posterior distributions of the model parameters. Interpret the results.

Answer:

Posterior density plots for the model parameters β_0 and β_1 are visualized in Figure 6. Both densities are unimodal and approximately symmetric, closely resembling normal distributions. Such symmetry suggests that the parameters are well identified by the data, with no significant skewness. This characteristic allows us to summarize the uncertainty effectively using the posterior mean and credible intervals, making the inference more straightforward and reliable. These summary statistics are presented in Table 2.

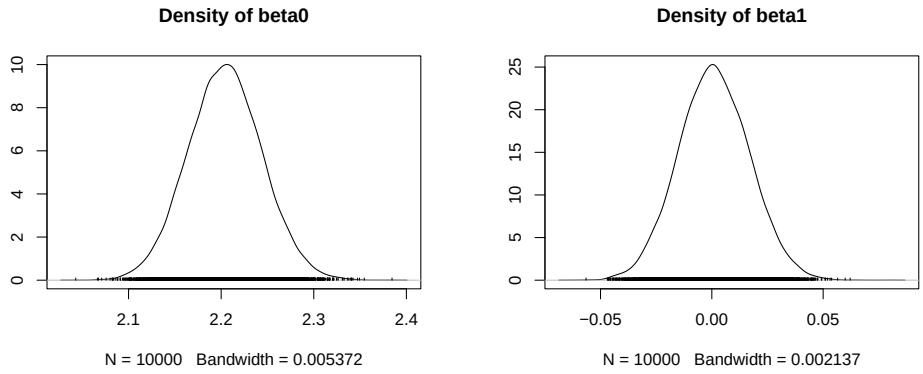


Figure 6: Density functions for β_0 and β_1 parameters

The results show the posterior mean of β_0 is approximately 2.20, with a small standard deviation of 0.039, indicating a precise estimate of the intercept. For β_1, the posterior mean is close to zero (0.001) with a standard deviation of 0.016, and the 95% Credible Interval ranges from slightly negative to slightly positive values, indicating considerable uncertainty about the effect of birth year on vaccination coverage.

Table 2: Posterior estimates and 95% Credible Intervals for parameters β_0 and β_1

Parameter	**Mean**	**SD**	**95% Credible Intervals**
β_0	2.204	0.039	[2.125 - 2.282]
β_1	0.001	0.016	[-0.029 - 0.0324]

While the intercept β_0 is estimated precisely, the slope parameter β_1 shows no strong evidence of a consistent positive or negative trend in vaccination coverage over birth years 2011–2019, since the

credible interval for β_1 includes zero. This suggests that, based on the data and the prior, zero is a credible (plausible) value for the effect size, meaning that there is substantial posterior probability that there is no change in vaccination coverage across birth cohorts.

Question 5:

Give the posterior estimate of the vaccination coverage per birth year. Compare with the analytical results you obtained in Question 1.

Answer:

The posterior estimates used in this question were obtained from the Bayesian logistic regression model established in Question 2. In this model, the vaccination rate (posterior estimate) was calculated for each birth year by taking the average MCMC samples of the β_0 and β_1 parameters.The probability was obtained by applying the inverse logit transformation to the linear predictor as follows:

$$\pi_{\text{year}} = \text{logit}^{-1}\left(\beta_0 + \beta_1 \cdot (\text{year} - 2015)\right)$$

This model accounts for the effect of birth year by centering it around 2015.

In question 1, there are two different analytical estimates obtained using conjugate beta distributions: one without prior information (flat prior: $\alpha = 1$, $\beta = 1$), and the other informative prior ($\alpha = 50$, $\beta = 2$) reflecting the expectation of a high vaccination rate in advance.

The results of these three approaches were compared and presented in Table 3, which summarizes the vaccination coverage estimates per birth year. When the results of this model are examined, it is observed that the predicted vaccination coverage remains approximately constant at around 90% across all birth years. This suggests that the model did not detect a meaningful trend over time. Accordingly, the slope parameter β_1, which represents the effect of birth year, is likely to be close to zero. More fluctuations are observed in analytical estimates made with a flat prior (e.g. 2018: 0.8733, 2019: 0.9133). Since this method is based only on data, if the data is low, the estimate is also low, causing year-based variability.

Table 3: Posterior and Analytical Vaccination Coverage Estimates (2011–2019)

Year	Logistic_Mean	Analytical_Mean_Flat	Analytical_Mean_Info
2011	0.9003	0.8900	0.9067
2012	0.9004	0.9067	0.9200
2013	0.9004	0.9033	0.9167
2014	0.9005	0.8833	0.9000
2015	0.9006	0.8900	0.9000
2016	0.9007	0.9067	0.9200
2017	0.9008	0.9000	0.9133
2018	0.9009	0.8733	0.8933
2019	0.9010	0.9133	0.9200

In analytical estimates made with an informative prior, the highest rates were generally obtained (e.g. 2012: 0.9200, 2019: 0.9200). This shows that the model takes into account the expectation of a high vaccination rate as prior information and pulls the results up.

Question 6:

Secondly, investigate whether the vaccination coverage trends are distinct at the different locations by adding a location-specific intercept and slope:

$$\text{logit}(\pi_{ij}) = \beta_{0i} + \beta_{1i} \cdot (\text{Year}_j - 2015)$$

Use data from the years 2011–2019. Assume non-informative priors for the parameters to be estimated. Write the code in BUGS language. Give a brief summary of the convergence checks you performed. Give the posterior estimates of this model.

Answer:

In this question, the given model uses index i to represent the region and index j to represent the year. As a result, the model estimates three separate pairs of coefficients: one intercept (β_0) and one slope (β_1) for each region.

The priors used for these parameters were vague (non-informative): $\beta_{0i}, \beta_{1i} \sim \mathcal{N}(0, 10^6)$. These wide normal distributions reflect no strong prior beliefs, allowing the posterior distributions to be mostly influenced by the observed data. As a result, the model estimates three separate pairs of coefficients: one intercept (β_0) and one slope (β_1) for each region.To check the convergence of chains, trace plots, auto-correlation plots, and Gelman-Rubin diagnosis plots are used for each parameter.

The BUGS model defines a Bayesian logistic regression with region-specific intercepts and slopes.

The likelihood assumes that the number of vaccinated individuals in each region–year combination follows a binomial distribution, with the probability of vaccination linked to year through a logistic function. The model includes three different β_{0i} and β_{1i} parameters for the three regions, accounting for distinct baseline vaccination rates and temporal trends. Non-informative normal priors with large variance ($\mathcal{N}(0, 10^6)$) were used for all parameters to reflect vague prior knowledge and allow the data to dominate the inference.

model BUGS

```
model {
  for (n in 1:Nobs) {
    # 1) Likelihood
    Y[n] ~ dbin(p[n], N[n])
    # 2) Logit link
    logit(p[n]) <- beta0[Region[n]] + beta1[Region[n]] * (Year[n] - 2015)
  }

  # Priors
  for (r in 1:Nregion) {
    beta0[r] ~ dnorm(0, 1.0E-6)
    beta1[r] ~ dnorm(0, 1.0E-6)
  }
}
```

Convergence Diagnostics and Interpretation:

To assess the convergence of the MCMC chains, trace plots and autocorrelation plots were examined. Figure 7 and Figure 8 displays the trace and density plots for the β_{1i} and β_{0i} parameters. The chains show stable fluctuations over the iterations, with no apparent trends or divergences. This indicates that the chains are well mixed and have likely reached stability.

Figure 9 presents the autocorrelation plots for all parameters. It can be seen that autocorrelation values beyond lag = 1 drop rapidly to near zero for all parameters. This suggests that the samples are approximately independent and that the chains mix efficiently. The combination of stable trace plots and rapidly declining autocorrelations supports the conclusion that the MCMC algorithm has converged and that the posterior estimates are reliable.

As shown in Figure 10, the Gelman-Rubin diagnostic plots indicate that the shrink factor values for all parameters rapidly approach the level of 1.0 as the number of iterations increases and remain stable at that level. This provides strong evidence that the MCMC sampling chains have successfully converged.

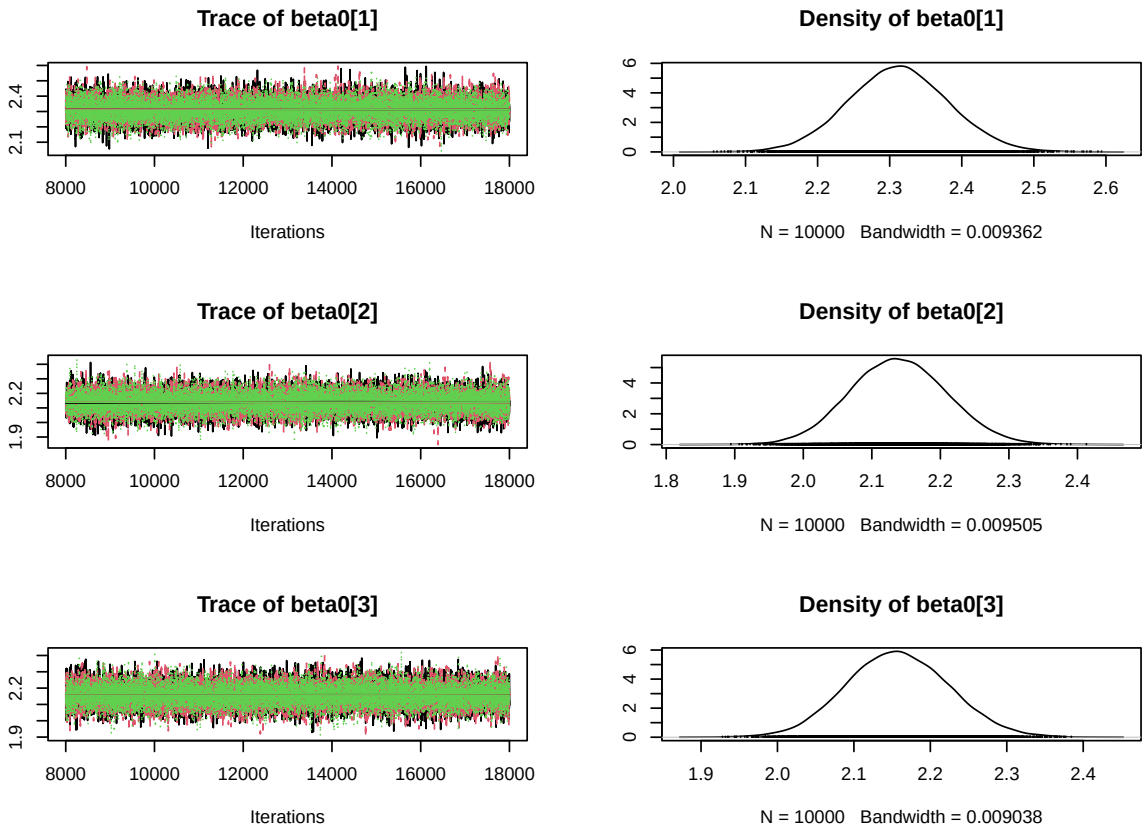


Figure 7: Trace and density plots for β_0 parameters

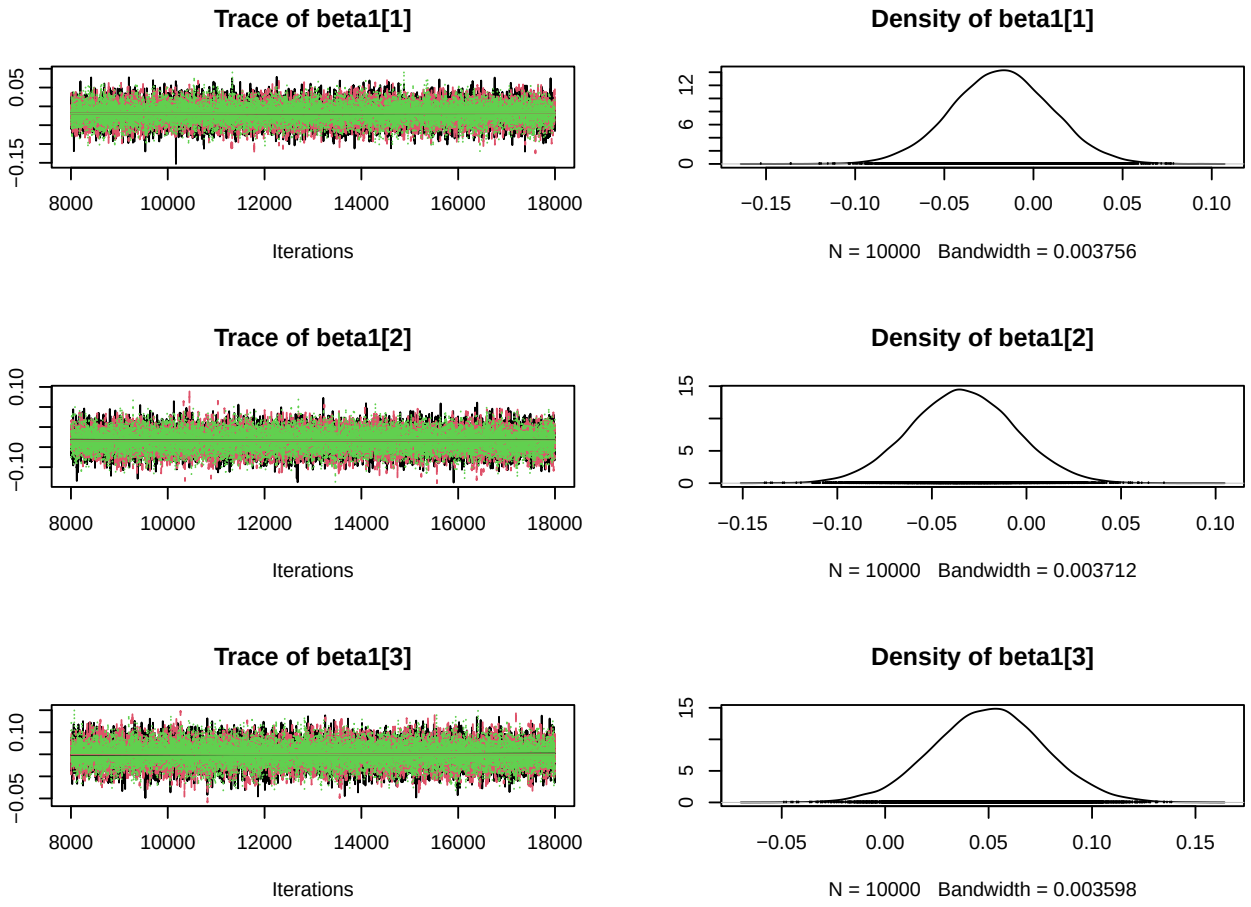


Figure 8: Trace and density plots for β_1 parameters

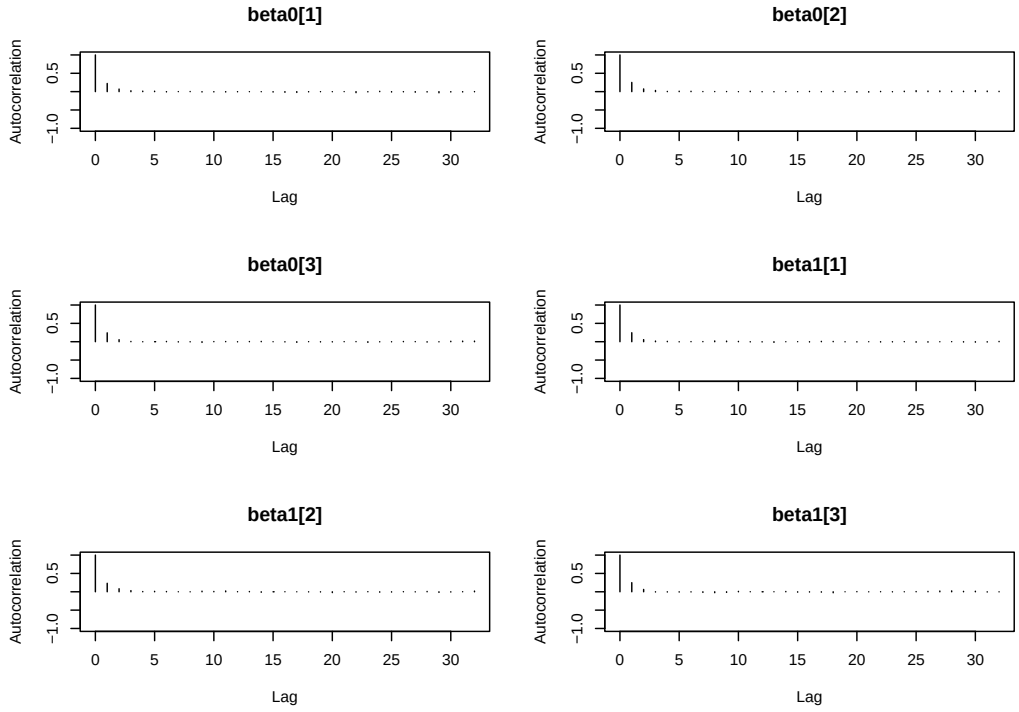


Figure 9: Autocorrelation plots for all parameters

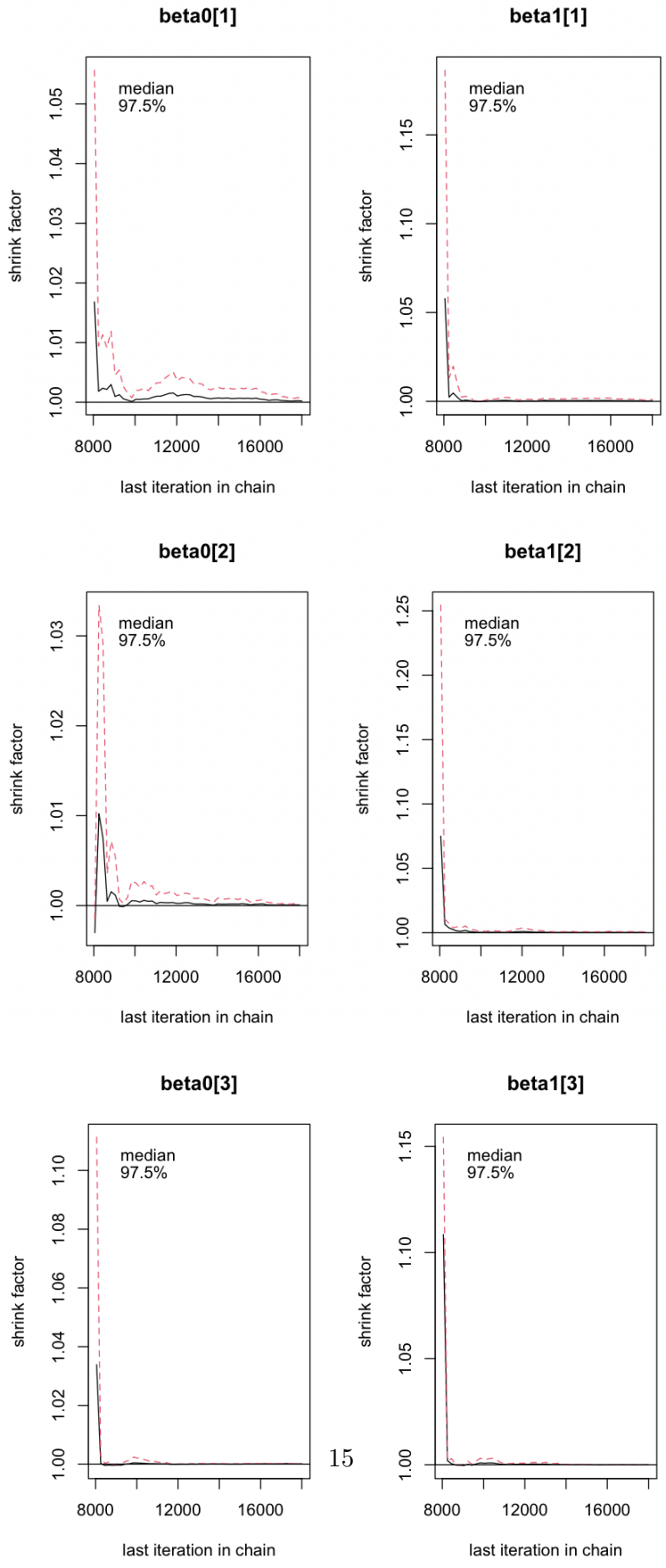


Figure 10: Gelman-Rubin shrink factor plots for each region's β_0 and β_1 parameters.

Interpretation of Posterior Estimates

As shown in Table 4, the posterior mean of the intercept parameter (β_0) is estimated precisely for each region, with small standard deviations and narrow 95% credible intervals that do not include zero. This indicates strong evidence for high baseline log-odds of vaccination coverage in the reference year across all regions. As shown in Table 4, for Georgia, the posterior distribution of $\beta_0[1]$ is centered at a relatively high value (mean = 2.31), indicating a high baseline (reference year) log-odds of vaccination coverage in this region. For Wisconsin and Mississippi, the posterior distributions of β_0 are also centered at high values, suggesting similarly high baseline vaccination coverage in these regions.

Posterior summary statistics for the region-specific slope parameters (β_1) are presented in Table 4. For all three regions, the 95% highest posterior density (HPD) intervals include zero. In a Bayesian context, this indicates that, given the data and the model, the posterior distribution of each slope parameter assigns some probability to values around zero. For Georgia and Wisconsin, the posterior distributions are centered near slightly negative values, suggesting a potential decreasing trend in vaccination coverage over time in these regions. For Mississippi, the posterior distribution of $\beta_1[3]$ is shifted toward positive values, with a mean of 0.05036, indicating a possible increasing trend.

However, since the 95% credible intervals include zero, we do not conclude that there is a definitive trend in any direction. Rather, the posterior distributions reflect uncertainty and allow for a range of plausible values, including the possibility of no change. This interpretation aligns with the Bayesian perspective, where credible intervals represent the degree of belief about parameter values given the observed data.

Table 4: Posterior estimates and 95% credible intervals for intercept (β_0) and slope (β_1) parameters by region

Region	**Parameter**	**Mean**	**SD**	**2.5%**	**97.5%**	**Includes 0?**
Georgia	β_0	2.31240	0.06959	2.17716	2.45027	No
	β_1	-0.01796	0.02785	-0.07316	0.03566	Yes
Wisconsin	β_0	2.13915	0.07048	2.00327	2.27946	No
	β_1	-0.03381	0.02752	-0.08763	0.02037	Yes
Mississippi	β_0	2.15766	0.06702	2.02902	2.29075	No
	β_1	0.05036	0.02672	-0.00211	0.10302	Yes

Question 7:

What is the probability (a posteriori) that there is an increase in vaccination coverage (per location)?

Answer:
After convergence checks were performed to confirm that the Bayesian estimates were reliable, MCMC samples for each β_{1i}, the slope per region, were extracted and converted to a matrix to allow access to these samples easily by parameter name. The posterior probability that the vaccination coverage increased over birth years (2011–2019) in each region was then computed. It means the posterior probability that β_{1i} is greater than zero for each region. Since β_{1i} represents the trend (slope) in vaccination coverage over time in region i, a positive value of β_{1i} indicates an increasing trend in coverage.

Table 5: Posterior probabilities per location.

Location	**Posterior** $P(\beta_1 > 0)$
Georgia	0.2629
Wisconsin	0.1123
Mississippi	0.9717

As you can see in the Table 5, there is strong evidence for an increasing vaccination trend (positive trend) in Mississippi (97.1%). In contrast, there is little to no support for a positive trend in Georgia (∼26%) or Wisconsin (∼11%), suggesting that vaccination coverage in these regions has likely remained stable or decreased slightly over the years 2011–2019.

Question 8:

Make a plot of the estimated vaccination coverage (per location and birth year), including the uncertainty on the estimates. Include also the observed vaccination proportion in the plot.

Answer:
Non-informative priors for intercepts and slopes were assumed, allowing the data to drive inference, and also posterior estimates were obtained using MCMC sampling, with convergence assessed through trace plots and Gelman-Rubin diagnostics. Having said that, Year is centered at 2015 to improve interpretability and MCMC convergence.
First, the data were prepared to include only years 2011 to 2019, and for each record, the observed vaccination proportion was calculated. Also, regions were encoded as numeric factors for easier indexing in the posterior samples. After that, using the MCMC posterior samples of the regression coefficients β_{0i} and β_{1i} for each region i, the posterior distribution of the vaccination coverage

probabilities for each year y was calculated by applying the inverse logit transformation to the linear predictor as follows:

$$\pi_{iy} = \text{logit}^{-1}\left(\beta_{0i} + \beta_{1i} \cdot (y - 2015)\right)$$

where π_{iy} is the vaccination probability in region i and year y.
For each region-year combination, the posterior coverage distribution was summarized by computing the posterior mean (expected coverage) and the 95% credible interval (2.5th and 97.5th percentiles). These posterior means over time (from 2011 to 2019) across three regions are plotted in Figure 11 with a line representing the mean estimated vaccination coverage, the shaded areas representing the 95% credible interval (uncertainty), and points are observed proportions.

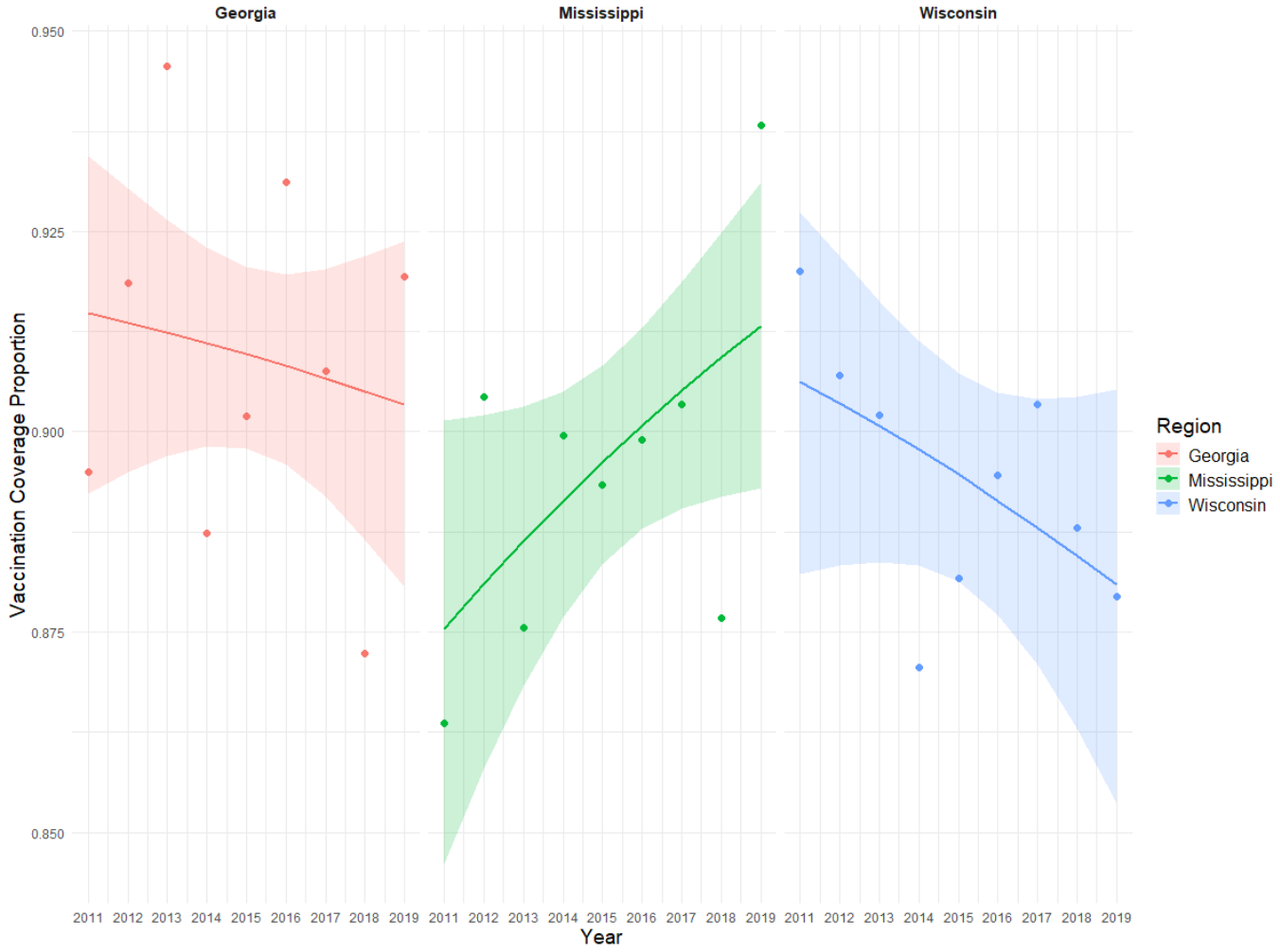


Figure 11: Estimated vaccination coverage by region and birth Year

The estimated trends reveal clear differences in both the slopes (rates of change) and baseline coverage levels across regions. Most observed data points fall well within the model's credible intervals, suggesting that the model adequately captures the variability in the data and fits the observed coverage well. In Georgia, there is a slight downward trend in vaccination coverage over time. In Wisconsin, the data suggest a more pronounced decrease, although the uncertainty is higher. In contrast, Mississippi shows strong evidence of an increasing trend in coverage, with most observed values closely matching the model's estimates. This suggests a consistent improvement in vaccination rates year over year in that region.
Overall, the model effectively captures regional variation in vaccination trends, and the credible intervals provide valuable insight into the uncertainty around these estimates.

Question 9:

Investigate whether the observed number of vaccinated children in 2020 is in line with the expectations from earlier years. For this, compare the observed number of vaccinated children in 2020 with the prediction intervals for number of vaccinated children in 2020.

Answer:
To determine whether the observed number of vaccinated children in 2020 is consistent with earlier years, we used a posterior predictive checking approach based on the Bayesian logistic regression model fitted in Question 6. Using the posterior samples of β_{0i} and β_{1i} obtained via MCMC, we computed predictive distributions for the number of vaccinated children in 2020 for each region i.

1. For each MCMC sample s, we computed the predicted log-odds of vaccination in 2020:

$$\text{logit}(\pi_{i,2020}^{(s)}) = \beta_{0i}^{(s)} + \beta_{1i}^{(s)} \cdot (2020 - 2015)$$

2. We transformed this to get the vaccination probability:

$$\pi_{i,2020}^{(s)} = \text{logit}^{-1}\left(\text{logit}\left(\pi_{i,2020}^{(s)}\right)\right)$$

3. We then simulated the number of vaccinated children:

$$\tilde{Y}_{i,2020}^{(s)} \sim \text{Binomial}(N_{i,2020}, \pi_{i,2020}^{(s)})$$

   where $N_{i,2020}$ is the observed sample size in 2020 for region i.

4. From the simulated $\tilde{Y}_{i,2020}^{(s)}$, we calculated summary statistics: the posterior predictive median and the 95% credible interval.

Table 6: Posterior Predictive Counts vs Observed Counts for 2020

Region	Observed Counts	Predicted Median	95% CI Lower	95% CI Upper
Georgia	165	170	160	178
Wisconsin	156	155	144	165
Mississippi	161	166	157	174

Results: The Table 6 compares the observed number of vaccinated children in 2020 with the predicted values from the posterior predictive distribution. In all three regions, the observed counts fall within the 95% credible intervals of the predicted distributions. This suggests that the 2020 data is consistent with what we would expect based on vaccination patterns from 2011 to 2019. This suggests that the Bayesian hierarchical model, trained on data from earlier years, fits the data well and provides reliable predictions for future outcomes.

Question 10:

Make pairwise comparisons of the vaccination coverage in 2019 by estimating the ratio of the vaccination coverage in 2019 in two locations. Interpret the results.

Answer:

To compare regional vaccination coverage in 2019, we used the posterior samples from the Bayesian logistic regression model fitted in Question 6. For each region i, we computed the posterior distribution of the vaccination probability in 2019 as:

$$\pi_{i,2019}^{(s)} = \text{logit}^{-1}\left(\beta_{0i}^{(s)} + \beta_{1i}^{(s)} \cdot (2019 - 2015)\right),$$

where $\beta_{0i}^{(s)}$ and $\beta_{1i}^{(s)}$ are the region-specific intercept and slope parameters from MCMC sample s, and the year was centered at 2015.

Using these samples, we then calculated the posterior distribution of the pairwise ratios of vaccination coverage between regions. For instance, to compare Georgia and Wisconsin, we computed:

$$R_{\text{GA/WI}}^{(s)} = \frac{\pi_{\text{GA},2019}^{(s)}}{\pi_{\text{WI},2019}^{(s)}}$$

A ratio greater than 1 in a given posterior sample implies that the numerator region had higher estimated coverage in that sample. Repeating this across all posterior results gives us both a distribution of the ratio and the posterior probability that one region had higher coverage than another.

Unlike frequentist methods, Bayesian analysis does not rely on p-values or null hypothesis rejection. Instead, we directly compute the probability that a hypothesis or condition is true given the

observed data. In this context, we estimate the posterior probability that one region had higher vaccination coverage than another in 2019 using:

$$\Pr\left(\frac{\pi_A}{\pi_B} > 1 \mid \text{data}\right) \approx \frac{1}{S}\sum_{s=1}^{S} \mathbb{I}\left(\frac{\pi_A^{(s)}}{\pi_B^{(s)}} > 1\right),$$

where $\mathbb{I}(\cdot)$ is the indicator function, and S is the total number of posterior samples. This gives a natural and interpretable way to quantify how confident we are that one region had higher coverage than another, without relying on significance testing.

Table 7: Posterior Distributions of 2019 Coverage Ratios

Comparison	**Posterior Mean**	**95% Credible Interval**	**Pr(Ratio > 1)**
Georgia / Wisconsin	1.026	[0.988, 1.067]	0.91
Georgia / Mississippi	0.989	[0.958, 1.021]	0.25
Wisconsin / Mississippi	0.964	[0.929, 0.999]	0.02

Results: Table 7 presents the summary statistics for each regional comparison:

- For Georgia vs. Wisconsin, the posterior mean ratio is 1.026, and the 95% credible interval ranges from 0.988 to 1.067. The posterior probability that Georgia had higher coverage than Wisconsin is 91%. This provides reasonably strong evidence favoring Georgia, though some uncertainty remains.
- In the comparison between Georgia and Mississippi, the mean ratio is 0.989, with a credible interval of [0.958, 1.021], and a 25% posterior probability that Georgia had higher coverage. This indicates a slight possible edge toward higher coverage in Mississippi, but the difference is highly uncertain and not clearly supported by the data.
- For Wisconsin vs. Mississippi, the ratio is 0.964 with a 95% credible interval from 0.929 to 0.999. The probability that Wisconsin had higher coverage is only 2%, indicating strong evidence that Mississippi likely had higher coverage in 2019.

Overall, while the differences in coverage are relatively small, the Bayesian posterior probabilities help quantify our confidence in which regions likely had higher vaccination rates in 2019.

Appendix - Tables

Table 8: Posterior summaries for vaccination coverage using informative and uninformative Beta priors

Region	Year	Prior Type	Posterior mean	Posterior variance	95% CI lower	95% CI upper
Wisconsin	2011	Informative Beta(50,2)	0.930	0.000241	0.895	0.955
Wisconsin	2011	Uninformative Beta(1,1)	0.920	0.000336	0.877	0.949
Wisconsin	2012	Informative Beta(50,2)	0.916	0.000272	0.882	0.947
Wisconsin	2012	Uninformative Beta(1,1)	0.907	0.000381	0.862	0.938
Wisconsin	2013	Informative Beta(50,2)	0.910	0.000276	0.878	0.943
Wisconsin	2013	Uninformative Beta(1,1)	0.902	0.000382	0.857	0.934
Wisconsin	2014	Informative Beta(50,2)	0.890	0.000360	0.848	0.922
Wisconsin	2014	Uninformative Beta(1,1)	0.870	0.000507	0.820	0.908
Wisconsin	2015	Informative Beta(50,2)	0.895	0.000299	0.859	0.926
Wisconsin	2015	Uninformative Beta(1,1)	0.882	0.000402	0.837	0.915
Wisconsin	2016	Informative Beta(50,2)	0.910	0.000262	0.871	0.934
Wisconsin	2016	Uninformative Beta(1,1)	0.894	0.000347	0.853	0.925
Wisconsin	2017	Informative Beta(50,2)	0.910	0.000271	0.879	0.943
Wisconsin	2017	Uninformative Beta(1,1)	0.905	0.000373	0.859	0.935
Wisconsin	2018	Informative Beta(50,2)	0.900	0.000303	0.864	0.932
Wisconsin	2018	Uninformative Beta(1,1)	0.881	0.000418	0.842	0.922
Wisconsin	2019	Informative Beta(50,2)	0.891	0.000339	0.856	0.928
Wisconsin	2019	Uninformative Beta(1,1)	0.881	0.000478	0.830	0.916
Wisconsin	2020	Informative Beta(50,2)	0.900	0.000393	0.858	0.935
Wisconsin	2020	Uninformative Beta(1,1)	0.881	0.000599	0.825	0.921
Mississippi	2011	Informative Beta(50,2)	0.884	0.000409	0.842	0.921
Mississippi	2011	Uninformative Beta(1,1)	0.864	0.000599	0.809	0.904
Mississippi	2012	Informative Beta(50,2)	0.910	0.000275	0.880	0.945
Mississippi	2012	Uninformative Beta(1,1)	0.904	0.000383	0.859	0.936
Mississippi	2013	Informative Beta(50,2)	0.890	0.000356	0.853	0.926
Mississippi	2013	Uninformative Beta(1,1)	0.876	0.000507	0.825	0.913
Mississippi	2014	Informative Beta(50,2)	0.910	0.000279	0.875	0.941
Mississippi	2014	Uninformative Beta(1,1)	0.900	0.000384	0.855	0.931
Mississippi	2015	Informative Beta(50,2)	0.901	0.000266	0.870	0.934
Mississippi	2015	Uninformative Beta(1,1)	0.893	0.000355	0.851	0.925
Mississippi	2016	Informative Beta(50,2)	0.910	0.000232	0.876	0.936
Mississippi	2016	Uninformative Beta(1,1)	0.899	0.000299	0.860	0.928
Mississippi	2017	Informative Beta(50,2)	0.910	0.000216	0.881	0.938
Mississippi	2017	Uninformative Beta(1,1)	0.903	0.000275	0.866	0.931
Mississippi	2018	Informative Beta(50,2)	0.890	0.000297	0.854	0.922
Mississippi	2018	Uninformative Beta(1,1)	0.875	0.000394	0.833	0.910
Mississippi	2019	Informative Beta(50,2)	0.940	0.000146	0.916	0.963
Mississippi	2019	Uninformative Beta(1,1)	0.939	0.000184	0.907	0.960
Mississippi	2020	Informative Beta(50,2)	0.910	0.000365	0.865	0.940
Mississippi	2020	Uninformative Beta(1,1)	0.889	0.000552	0.835	0.927

Appendix - R code

```r
library(rjags)
library(runjags)
library(coda)
library(ggplot2)
library(readxl)
library(coda)
library(dplyr)
library(mcmcplots)

# Question 1
data <- data.frame(
  Region = rep(c("Georgia", "Wisconsin", "Mississippi"), each = 10),
  Year = rep(2011:2020, times = 3),
  Vaccinated = c(
    196,248,261,252,276,311,265,246,251,165,      # Georgia
    207,205,212,195,231,246,215,214,197,156,      # Wisconsin
    171,208,190,215,243,276,290,242,304,161       # Mississippi
  ),
  Sample_Size = c(
    219,270,276,284,306,334,292,282,273,188,      # Georgia
    225,226,235,224,262,275,238,241,224,177,      # Wisconsin
    198,230,217,239,272,307,321,276,324,181       # Mississippi
  )
)

flat_results <- data %>%
  mutate(
    post_alpha   = 1 + Vaccinated,
    post_beta    = 1 + (Sample_Size - Vaccinated),
    mean_flat    = (post_alpha / (post_alpha + post_beta)),
    mean_flat2 = round(mean_flat, 2),
    var_flat = (post_alpha*post_beta)/ (((post_alpha+post_beta)^2)*(post_alpha+post_beta+1)
        ),
    ci_flat_low  = qbeta(0.025, post_alpha, post_beta),
    ci_flat_high = qbeta(0.975, post_alpha, post_beta)
  )
inf_results <- data %>%
  mutate(
    post_alpha   = 50 + Vaccinated,
    post_beta    = 2 + (Sample_Size - Vaccinated),
    mean_inf     = post_alpha / (post_alpha + post_beta),
    mean_inf_2 = round(mean_inf, 2),
    var_inform = (post_alpha*post_beta)/ (((post_alpha+post_beta)^2)*(post_alpha+post_beta
        +1)),
    ci_inf_low   = qbeta(0.025, post_alpha, post_beta),
    ci_inf_high  = qbeta(0.975, post_alpha, post_beta)
  )

all_results <- flat_results %>%
  select(Region,Year,Vaccinated,Sample_Size, mean_flat2,var_flat, ci_flat_low, ci_flat_high
      ) %>%
  left_join(
    inf_results %>%
      select(Region,Year, mean_inf_2, var_inform, ci_inf_low, ci_inf_high),
```

```r
    by = c("Region","Year")
  )
View(all_results)
print(all_results)

# Question 2

# filtering years making it from 2011 to 2019
data_q2 <- data %>%
  filter(Year >= 2011, Year <= 2019)
dataList <- list(
  Y    = data_q2$Vaccinated,
  N    = data_q2$Sample_Size,
  Year = data_q2$Year,
  Nobs = nrow(data_q2)
)

model.inits <- list(
  list(beta0 = -1, beta1 = 0, .RNG.name = "base::Wichmann-Hill", .RNG.seed = 1001),
  list(beta0 = 0,   beta1 = 0,   .RNG.name = "base::Wichmann-Hill", .RNG.seed = 1002),
  list(beta0 = 1,   beta1 = 0,.RNG.name = "base::Wichmann-Hill", .RNG.seed = 1003)
)

modelString <- "
model {
  for (n in 1:Nobs) {
    # 1) Likelihood
    Y[n] ~ dbin(pi[n], N[n])
    # 2) Logit link
    logit(pi[n]) <- beta0 + beta1 * (Year[n] - 2015)
  }

  # Priors
  beta0 ~ dnorm(0, 1.0E-6)
  beta1 ~ dnorm(0, 1.0E-6)
}
"
jags <- jags.model(textConnection(modelString),
                   data = dataList,
                   inits = model.inits,
                   n.chains = 3)
# Question 3
# run  or generate MCMC samples
# the burn-in
update(jags, n.iter=7000)
# Draw samples
samples <- coda.samples(
  jags,
  variable.names=c("beta0","beta1"),
  n.iter=10000, thin=1
)
# History plot & posterior distributions & autocorrelation plot
plot(samples, trace=TRUE, density = TRUE)
sample.mcmc <- as.mcmc.list(samples)
# Checking correlations
autocorr.plot(sample.mcmc)
autocorr.plot(samples)
```

```r
# Convergence tests
gelman.diag(sample.mcmc)
gelman.plot(sample.mcmc,ask=FALSE)

# Question 4
# Combine multiple chains to get overall HPD interval
# WE CAN HAVE POSTERIOR SUMMARY AND DENSITY PLOT FROM HERE
library(runjags)
model.mcmc.combined<-combine.mcmc(samples)
View(model.mcmc.combined)
HPDinterval(model.mcmc.combined)
summary(model.mcmc.combined)
plot(model.mcmc.combined, trace=TRUE, density = TRUE)
plot(samples, trace=TRUE, density = TRUE)

#question5

post_coefs <- colMeans(as.matrix(samples)[, c("beta0","beta1")])
years <- 2011:2019

logistic_means <- plogis(
  post_coefs["beta0"] + post_coefs["beta1"] * (years - 2015)
)

analytical_means_flat <- flat_results %>%
  filter(Year %in% years) %>%
  group_by(Year) %>%
  summarize(analytic_mean = mean(mean_flat2)) %>%
  pull(analytic_mean)


analytical_means_info <- inf_results %>%
  filter(Year %in% years) %>%
  group_by(Year) %>%
  summarize(analytic_mean = mean(mean_inf_2)) %>%
  pull(analytic_mean)

compare_df <- data.frame(
  Year               = years,
  Logistic_Mean      = round(logistic_means,    4),
  Analytical_Mean_Flat    = round(analytical_means_flat,  4),
  Analytical_Mean_info    = round(analytical_means_info,  4)
)

#Question 6
data_q2 <- data %>%
  filter(Year >= 2011, Year <= 2019)
dataList2 <- list(
  Y    = data_q2$Vaccinated,
  N    = data_q2$Sample_Size,
  Year = data_q2$Year,
  Region = as.numeric(factor(data_q2$Region, levels = c("Georgia", "Wisconsin", "
      Mississippi"))),
  Nobs = nrow(data_q2),
  Nregion = length(unique(data_q2$Region))
)
model.inits2 <- function() {
```

```
  list(
    beta0 = rnorm(3, 0, 0.1),
    beta1 = rnorm(3, 0, 0.1)
  )
}

modelString2 <- "
model {
  for (n in 1:Nobs) {
    # 1) Likelihood
    Y[n] ~ dbin(p[n], N[n])
    # 2) Logit link
    logit(p[n]) <- beta0[Region[n]] + beta1[Region[n]] * (Year[n] - 2015)
  }

  # Priors
  for (r in 1:Nregion) {
    beta0[r] ~ dnorm(0, 1.0E-6)
    beta1[r] ~ dnorm(0, 1.0E-6)
  }
}
"
set.seed(12345)
jags2 <- jags.model(textConnection(modelString2),
                    data = dataList2,
                    inits = model.inits2,
                    n.chains = 3)

update(jags2, n.iter=7000)

samples2 <- coda.samples(
  jags2,
  variable.names=c("beta0","beta1"),
  n.iter=10000, thin=1
)

plot(samples2, trace=TRUE, density = TRUE)

sample2.mcmc <- as.mcmc.list(samples2)
gelman.diag(sample2.mcmc)
gelman.plot(sample2.mcmc,ask=FALSE)
autocorr.plot(sample2.mcmc)

model.mcmc.combined <- combine.mcmc(samples2)
HPDinterval(model.mcmc.combined)
summary(sample2.mcmc)


#Question 7
posterior_samples <- as.matrix(samples2)
# For each location, compute P(beta1[i] > 0)
prob_increase <- c(
  Georgia = mean(posterior_samples[, "beta1[1]"] > 0),
  Wisconsin = mean(posterior_samples[, "beta1[2]"] > 0),
  Mississippi = mean(posterior_samples[, "beta1[3]"] > 0)
)
round(prob_increase, 4)
```

```r


# Question 8
observed_coverage <- data %>%
  filter(Year >= 2011 & Year <= 2019) %>%
  mutate(
    regions = as.numeric(factor(Region, levels = c("Georgia", "Wisconsin", "Mississippi"))),
    Observed = Vaccinated / Sample_Size
  )

years <- unique(observed_coverage$Year)

coverage_preds <- do.call(rbind, lapply(1:3, function(i) {
  sapply(years, function(y) {
    logit <- posterior_samples[, paste0("beta0[", i, "]")] + posterior_samples[, paste0("beta1[", i, "]")] * (y - 2015)
    p <- 1 / (1 + exp(-logit))
    c(mean = mean(p), lower = quantile(p, 0.025), upper = quantile(p, 0.975))
  }) %>%
    t() %>%
    as.data.frame() %>%
    mutate(Year = years, Region = c("Georgia", "Wisconsin", "Mississippi")[i])
}))

ggplot(coverage_preds, aes(x = Year, y = mean, color = Region, fill = Region)) +
  geom_line(size = 1) +
  geom_ribbon(aes(ymin = `lower.2.5%`, ymax = `upper.97.5%`), alpha = 0.2, color = NA) +
  geom_point(data = observed_coverage, aes(x = Year, y = Observed, color = Region), size = 2, shape = 21) +
  facet_wrap(~Region) +
  scale_x_continuous(breaks = 2011:2019) +
  labs(y = "Vaccination Coverage Proportion", x = "Year") +
  theme_minimal()+ theme(
    axis.title.x = element_text(size = 14),
    axis.title.y = element_text(size = 14),
    legend.title = element_text(size = 14),
    legend.text = element_text(size = 12),
    strip.text.x = element_text(size = 11, face = "bold")
)

# Question 9
# Define 2020 sample sizes for each region
sample_sizes_2020 <- c(Georgia = 188, Wisconsin = 177, Mississippi = 181)
observed_2020 <- c(Georgia = 165, Wisconsin = 156, Mississippi = 161)

# Region names
regions <- c("Georgia", "Wisconsin", "Mississippi")

# Run posterior predictive simulation for each region
predictive_results <- lapply(1:3, function(i) {
  region_name <- regions[i]
  size_2020 <- sample_sizes_2020[region_name]
  obs_2020 <- observed_2020[region_name]

  # Get posterior predictive probabilities
  logit_p <- posterior_samples[, paste0("beta0[", i, "]")] +
```

```r
    posterior_samples[, paste0("beta1[", i, "]")] * (2020 - 2015)
  p <- 1 / (1 + exp(-logit_p))

  # Simulate vaccinated counts
  y_pred <- rbinom(length(p), size = size_2020, prob = p)

  # Summary statistics
  data.frame(
    Region = region_name,
    Observed = obs_2020,
    Mean = median(y_pred),
    Lower = quantile(y_pred, 0.025),
    Upper = quantile(y_pred, 0.975),
    stringsAsFactors = FALSE
  )
})

# Combine into a data frame
pred_summary <- do.call(rbind, predictive_results)
print(pred_summary)


# Questin 10
invlogit <- function(x) 1 / (1 + exp(-x))
year_centered <- 2019 - 2015

# Compute posterior probabilities for 2019
p_G <- invlogit(posterior_samples[, "beta0[1]"] + (year_centered * posterior_samples[, "beta1[1]"]))  # Georgia
p_W <- invlogit(posterior_samples[, "beta0[2]"] + (year_centered * posterior_samples[, "beta1[2]"]))  # Mississippi
p_M <- invlogit(posterior_samples[, "beta0[3]"] +(year_centered * posterior_samples[, "beta1[3]"]))  # Wisconsin

# Ratios
Georgia_Wisconsin <- p_G / p_W
Georgia_Mississippi <- p_G / p_M
Wisconsin_Mississippi <- p_W / p_M

# Summarise
summary_stats <- function(x) {
  c(mean = mean(x),
    lower = quantile(x, 0.025),
    upper = quantile(x, 0.975))
}

results <- rbind(
  GA_vs_WI = summary_stats(Georgia_Wisconsin),
  GA_vs_MS = summary_stats(Georgia_Mississippi),
  WI_vs_MS = summary_stats(Wisconsin_Mississippi)
)

round(results, 3)


# Compute posterior probabilities that the ratio > 1
Pr_ratio_gt_1 <- c(
```

```
  GA_vs_WI = mean(Georgia_Wisconsin > 1),
  GA_vs_MS = mean(Georgia_Mississippi > 1),
  WI_vs_MS = mean(Wisconsin_Mississippi > 1)
)

# Display rounded results
round(Pr_ratio_gt_1, 2)
```